

HW0428_10.20

2026-05-05

```
# a.
capm5 <- read.csv("C:/Users/tiffa/Downloads/capm5.csv")

# 計算 Microsoft 股票的風險溢價 (msftrf) 和市場的風險溢價 (rmrf)
capm5$msftrf = (capm5$msft - capm5$riskfree)
capm5$rmrf = (capm5$mkt - capm5$riskfree)

# 執行 CAPM 回歸模型
msftrf = lm(msftrf ~ rmrf, data = capm5)

# 顯示回歸結果
summary(msftrf)
```

```
##
## Call:
## lm(formula = msftrf ~ rmrf, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27424 -0.04744 -0.00820  0.03869  0.35801
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003250   0.006036   0.538   0.591
## rmrf         1.201840   0.122152   9.839  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08083 on 178 degrees of freedom
## Multiple R-squared:  0.3523, Adjusted R-squared:  0.3486
## F-statistic: 96.8 on 1 and 178 DF, p-value: < 2.2e-16
```

```
# 計算 beta (rmrf) 的 95% 區間估計
beta_95ci <- confint(msftrf, level = 0.95)["rmrf", ]

cat("ANS a. ", "\n")
```

```
## ANS a.
```

```
cat("The 95% confidence interval for beta is [", round(beta_95ci[1], 4), ", ", round(beta_95ci[2], 4), "]\n")
```

```
## The 95% confidence interval for beta is [ 0.9608 , 1.4429 ]
```

```
cat("Microsoft 的 beta 值(即 rmrf)估計為1.2018,95%信賴水準下區間估計為[0.9607882, 1.442891]。因為大於1,Microsoft 是屬於風險高於市場投組的標的。", "\n")
```

```
## Microsoft 的 beta 值(即 rmrf)估計為1.2018,95%信賴水準下區間估計為[0.9607882, 1.442891]。因為大於1,Microsoft 是屬於風險高於市場投組的標的。
```

```
# b.  
capm5$rank = rank(capm5$rmrf)  
rmrf.ols=lm(rmrf~rank , data=capm5)  
summary(rmrf.ols)
```

```
##  
## Call:  
## lm(formula = rmrf ~ rank, data = capm5)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.110497 -0.006308  0.001497  0.009433  0.029513   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept) -7.903e-02  2.195e-03  -36.0   <2e-16 ***  
## rank         9.067e-04  2.104e-05   43.1   <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.01467 on 178 degrees of freedom  
## Multiple R-squared:  0.9126, Adjusted R-squared:  0.9121   
## F-statistic: 1858 on 1 and 178 DF,  p-value: < 2.2e-16
```

```
cat("ANS b. ", "\n")
```

```
## ANS b.
```

```
cat("R2 = 0.9126 ,RANK 的 t 值為 43.10,對應的 F 值是 1857.61。因此 RANK 是一個很強的工具變數。", "\n")
```

```
## R2 = 0.9126 ,RANK 的 t 值為 43.10,對應的 F 值是 1857.61。因此 RANK 是一個  
## 很強的工具變數。
```

```
# c.
# 重新計算第一階段的殘差
capm5$rank = rank(capm5$rmrf)
rmrf.ols = lm(rmrf ~ rank, data = capm5)

# 提取第一階段回歸的殘差 v_hat
capm5$v_hat = residuals(rmrf.ols)

# 現在將殘差添加到 CAPM 模型中，並執行回歸分析
capm5$msftrf = (capm5$msft - capm5$riskfree)
capm5$rmrf = (capm5$mkt - capm5$riskfree)
capm5$rank = rank(capm5$rmrf)

capm5_model = lm(msftrf ~ rmrf + v_hat, data = capm5)

# 顯示擴充回歸的結果
summary(capm5_model)
```

```
##
## Call:
## lm(formula = msftrf ~ rmrf + v_hat, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27140 -0.04213 -0.00911  0.03423  0.34887
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003018   0.005984   0.504   0.6146
## rmrf         1.278318   0.126749  10.085 <2e-16 ***
## v_hat        -0.874599   0.428626  -2.040  0.0428 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08012 on 177 degrees of freedom
## Multiple R-squared:  0.3672, Adjusted R-squared:  0.36
## F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

```
cat("ANS c. ", "\n")
```

```
## ANS c.
```

```
cat("使用對內生性的Hausman測試。v_hat的t-統計量為-2.04,對應的p值為0.043。
在 5% 顯著水準(p 值 < 0.05)下,v_hat 顯著。在 1% 顯著水準(p 值 < 0.01)下,v_hat不顯著,因為 p值大於
0.01。在 1%顯著水準下,我們無法拒絕市場回報外生性的虛無假設。", "\n")
```

```
## 使用對內生性的Hausman測試。v_hat的t-統計量為-2.04,對應的p值為0.043。
## 在 5% 顯著水準(p 值 < 0.05)下,v_hat 顯著。在 1% 顯著水準(p 值 < 0.01)下,v_hat不顯著,因為 p值大
於0.01。在 1%顯著水準下,我們無法拒絕市場回報外生性的虛無假設。
```

```
# d.  
# 使用 rank 作為工具變數，進行 IV/2SLS 估計  
#install.packages("AER")  
library(AER)
```

```
## Loading required package: car
```

```
## Loading required package: carData
```

```
## Loading required package: lmtest
```

```
## Loading required package: zoo
```

```
##  
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':  
##  
##      as.Date, as.Date.numeric
```

```
## Loading required package: sandwich
```

```
## Loading required package: survival
```

```
iv_model = ivreg(msftrf ~ rmrf | rank, data = capm5)  
  
# 顯示 IV 估計結果  
summary(iv_model)
```

```
##  
## Call:  
## ivreg(formula = msftrf ~ rmrf | rank, data = capm5)  
##  
## Residuals:  
##      Min      1Q   Median      3Q      Max   
## -0.271625 -0.049675 -0.009693  0.037683  0.355579   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)  0.003018   0.006044   0.499   0.618      
## rmrf         1.278318   0.128011   9.986  <2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.08092 on 178 degrees of freedom  
## Multiple R-Squared:  0.3508, Adjusted R-squared:  0.3472   
## Wald test: 99.72 on 1 and 178 DF, p-value: < 2.2e-16
```

```
# 計算 beta 的 95% 信賴區間
beta_95CI = confint(iv_model, level = 0.95)["rmrf", ]

cat("ANS d. ", "\n")
```

```
## ANS d.
```

```
cat("The 95% confidence interval for beta is : [", round(beta_95CI, 4), "]\n")
```

```
## The 95% confidence interval for beta is : [ 1.0274 1.5292 ]
```

```
cat("95%的信賴區間是 [1.0257, 1.5309]。所有數值都大於1,因此我們拒絕微軟的beta等於1 的假設。",
"\n")
```

```
## 95%的信賴區間是 [1.0257, 1.5309]。所有數值都大於1,因此我們拒絕微軟的beta等於1 的假設。
```

```
# e.
# 創建 POS 變數
capm5$POS <- ifelse(capm5$rmrf > 0, 1, 0)
# 使用 RANK 和 POS 作為工具變數，進行第一階段回歸
first_stage_model_e <- lm(rmrf ~ rank + POS, data = capm5)
# 顯示回歸結果
summary(first_stage_model_e)
```

```
##
## Call:
## lm(formula = rmrf ~ rank + POS, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.109182 -0.006732  0.002858  0.008936  0.026652
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.0804216  0.0022622  -35.55  <2e-16 ***
## rank         0.0009819  0.0000400   24.55  <2e-16 ***
## POS         -0.0092762  0.0042156   -2.20   0.0291 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.01451 on 177 degrees of freedom
## Multiple R-squared:  0.9149, Adjusted R-squared:  0.9139
## F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16
```

```
cat("ANS e. ", "\n")
```

```
## ANS e.
```

```
cat("F 統計量為 951.3 大於 10。RANK 仍然顯著,而 POS 則不顯著。p 值<0.01 ,拒絕誤差項與解釋變數之間不存在相關性的虛無假設。R2 = 0.9149,這些工具變數並不弱。", "\n")
```

```
## F 統計量為 951.3 大於 10。RANK 仍然顯著,而 POS 則不顯著。p 值<0.01 ,拒絕誤差項與解釋變數之間不存在相關性的虛無假設。R2 = 0.9149,這些工具變數並不弱。
```

```
# f.  
# 進行第一階段回歸，並提取殘差  
first_stage_model_e <- lm(rmrfr ~ rank + POS, data = capm5)  
capm5$residuals_first_stage <- residuals(first_stage_model_e)  
  
# 將第一階段的殘差加到 CAPM 模型中  
capm5_model <- lm(msftrf ~ rmrfr + residuals_first_stage, data = capm5)  
  
# 顯示回歸結果  
summary(capm5_model)
```

```
##  
## Call:  
## lm(formula = msftrf ~ rmrfr + residuals_first_stage, data = capm5)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -0.27132 -0.04261 -0.00812  0.03343  0.34867   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)    0.003004   0.005972   0.503   0.6157      
## rmrfr          1.283118   0.126344  10.156 <2e-16 ***  
## residuals_first_stage -0.954918  0.433062  -2.205  0.0287 *    
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 0.07996 on 177 degrees of freedom  
## Multiple R-squared:  0.3696, Adjusted R-squared:  0.3625   
## F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

```
# Hausman 測試，檢查市場回報是否外生  
hausman_test_result <- linearHypothesis(capm5_model, "residuals_first_stage = 0")  
print(hausman_test_result)
```

```
##
## Linear hypothesis test:
## residuals_first_stage = 0
##
## Model 1: restricted model
## Model 2: msftrf ~ rmrf + residuals_first_stage
##
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1     178 1.1629
## 2     177 1.1318  1    0.03109 4.8622 0.02874 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
cat("ANS f. ", "\n")
```

```
## ANS f.
```

```
cat("v_hat2(residuals_first_stage)的p值大於0.01,無法拒絕market return為外生性的假設。", "\n")
```

```
## v_hat2(residuals_first_stage)的p值大於0.01,無法拒絕market return為外生性的假設。
```

```
# g.
# 使用 RANK 和 POS 作為工具變數，進行 IV/2SLS 估計
library(AER) # 需要安裝 AER 套件來使用 ivreg 函數

# 設定工具變數，估算 IV/2SLS 模型
capm_iv_model <- ivreg(msftrf ~ rmrf | rank + POS, data = capm5)

# 顯示 IV 估計結果
summary(capm_iv_model)
```

```
##
## Call:
## ivreg(formula = msftrf ~ rmrf | rank + POS, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27168 -0.04960 -0.00983  0.03762  0.35543
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003004   0.006044   0.497    0.62
## rmrf         1.283118   0.127866  10.035 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08093 on 178 degrees of freedom
## Multiple R-Squared:  0.3507, Adjusted R-squared:  0.347
## Wald test: 100.7 on 1 and 178 DF, p-value: < 2.2e-16
```

```
# 估算OLS模型
```

```
ols_model <- lm(msftrf ~ rmrfr, data = capm5)
summary(ols_model)
```

```
##
## Call:
## lm(formula = msftrf ~ rmrfr, data = capm5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27424 -0.04744 -0.00820  0.03869  0.35801
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.003250   0.006036   0.538   0.591
## rmrfr        1.201840   0.122152   9.839 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.08083 on 178 degrees of freedom
## Multiple R-squared:  0.3523, Adjusted R-squared:  0.3486
## F-statistic: 96.8 on 1 and 178 DF,  p-value: < 2.2e-16
```

```
# 比較 IV 和 OLS 估計結果
```

```
iv_beta <- coef(capm_iv_model)["rmrfr"]
ols_beta <- coef(ols_model)["rmrfr"]
cat("IV Estimate of beta:", iv_beta, "\n")
```

```
## IV Estimate of beta: 1.283118
```

```
cat("OLS Estimate of beta:", ols_beta, "\n")
```

```
## OLS Estimate of beta: 1.20184
```

```
# 計算 rmrfr 的 95% 區間估計
```

```
iv_beta_95CI <- confint(capm_iv_model, level = 0.95)["rmrfr",]
```

```
cat("The 95% confidence interval for beta in the IV model is: [", iv_beta_95CI[1], ", ", iv_beta_95CI[2], "]\n")
```

```
## The 95% confidence interval for beta in the IV model is: [ 1.032505 , 1.533731 ]
```

```
cat("ANS g. ", "\n")
```

```
## ANS g.
```

```
cat("beta係數在2SLS模型中的估計值1.283118大於OLS模型中的估計值1.20184。如果存在測量誤差,則OLS估計量會受到衰減偏誤(偏向下估計)。故此2SLS模型中估計值略大是我們所預期的。其95%的區間估計為[1.032505 , 1.533731]。範圍都大於 1,因此拒絕Microsoft的beta等於1的虛無假設。", "\n")
```

```
## beta係數在2SLS模型中的估計值1.283118大於OLS模型中的估計值1.20184。如果存在測量誤差,則OLS估計量會受到衰減偏誤(偏向下估計)。故此2SLS模型中估計值略大是我們所預期的。其95%的區間估計為[1.032505 , 1.533731]。範圍都大於 1,因此拒絕Microsoft的beta等於1的虛無假設。
```

```
# h.  
# 提取 IV 回歸的殘差  
residuals_iv = residuals(capm_iv_model)  
  
# 進行 OLS 回歸,將 IV 回歸殘差作為因變數, RANK 和 POS 作為自變數  
OLS_eIV = lm(residuals_iv ~ rank + POS, data = capm5)  
  
# 修正重點:將 summary() 的結果存入變數中  
summary_OLS_eIV = summary(OLS_eIV)  
  
# 計算 R-squared 值並乘以樣本數  
N = length(capm5$msft) # 或者是數據中的其他觀察數量  
Rsquard = summary_OLS_eIV$r.squared  
Rsquard*N
```

```
## [1] 0.5584634
```

```
# 計算 Sargan test 的卡方臨界值  
chi_2 = qchisq(0.95, 1) # 假設顯著性水平為 5%  
chi_2_critical = chi_2  
  
# 輸出 Sargan 統計量與臨界值  
cat("Sargan test statistic (NR^2) is", round(Rsquard*N, 6), "\n")
```

```
## Sargan test statistic (NR^2) is 0.558463
```

```
cat("The 5% critical value for the Sargan test is", round(chi_2_critical, 3), "\n")
```

```
## The 5% critical value for the Sargan test is 3.841
```

```
cat("ANS h. ", "\n")
```

```
## ANS h.
```

```
# 根據 Sargan 統計量進行假設檢驗  
if (Rsquard*N > chi_2_critical) {  
  cat("We reject the null hypothesis that the surplus IV is valid.\n")  
} else {  
  cat("We fail to reject the null hypothesis that the surplus IV is valid.\n")  
}
```

We fail to reject the null hypothesis that the surplus IV is valid.